

Project Proposal & Implementation Roadmap: Do Circuits Survive Compression? A Null-Model Audit of Circuit Stability Under Quantization and Pruning

Target Venue: AAAI Undergraduate Consortium (AAAI UC)

Primary keywords: ML: Transparent, Interpretable & Explainable ML | ML: Efficient, Edge, Green & Hardware-aware ML

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Abstract

Interpretability findings are produced on full-precision models and consumed on compressed ones. Two recent audits have asked whether that transfer holds, but both study *representations*: sparse-autoencoder features under pruning and under quantization, and frozen linear probes under routine model updates. Neither studies *computation*. Circuits are the structural unit the field actually reasons with, and their behavior under quantization or weight pruning is essentially unmeasured. The difficulty is that circuit measurement carries two documented pathologies. Circuits reflect the analyst's pruning threshold as much as the model, and activation patching, the primary causal tool, estimates a quantity that mixes a component's own effect with interaction effects involving other components. Any claim that compression broke a circuit therefore needs a null model before it means anything. This project builds that null model first (matched-magnitude random weight perturbation, seed variation, threshold ensembles), then measures circuit change under real compression against that floor, reporting edge inclusion frequencies rather than binary membership and testing every conclusion at two levels of description. We additionally test whether circuit-level damage rankings agree with the feature-level damage rankings reported on the same models, which is the point at which a representation audit and a computation audit either corroborate or contradict each other.

1 Introduction & Motivation

Quantization and pruning are the standard route to deployment, and a compressed model is conventionally accepted when perplexity or downstream accuracy stays close to the full-precision original. Recent work has shown this acceptance criterion is not sufficient for interpretability. Borobia et al. [1] found that unstructured pruning reshapes SAE feature geometry, that rare low-firing features survive far better than frequent ones, and that geometric feature survival does not predict causal importance under ablation. Duan [2] showed that on Gemma-2-2B, INT7 quantization *improves* perplexity while degrading 18.7% of active SAE features, and that at INT6 only 51.3% of active features survive. A companion study [3] found that quantization-style updates largely preserve frozen probe performance while fine-tuning-style updates frequently make probes stale. The audit perspective is established; its coverage is not.

Every one of those studies measures a representation: a feature direction, a probe vector, a geometry. None measures a mechanism. This is a real gap, because the artifact that safety and interpretability work actually reasons about (“the IOI circuit”, “the refusal pathway”, “the induction heads”) is a computation graph, not a set of directions, and a model can preserve its features while rerouting the computation that consumes them, or vice versa. The single existing study that touches circuits under compression [4] reports, for vision-language models, that pruning largely leaves circuit structure intact while quantization modifies circuits at a higher level. That is a qualitative claim about a quantity whose noise floor has never been established.

Establishing that floor is not a formality; it is the hard part, and it is where the contribution lies. Circuit measurement inherits two pathologies that are documented, citable, and directly relevant:

- **Threshold dependence.** CIRCUS [5] reframes circuit discovery as uncertainty quantification over analyst degrees of freedom, showing that two pruning thresholds yield two different circuits with no principled way to say which edges reflect model structure. It assigns each edge a stability score, the fraction of pruning configurations that retain it, and finds that strict-consensus circuits on Gemma-2-2B and Llama-3.2-1B are roughly 40x smaller than the union of configurations while retaining comparable explanatory power. Independently, [6] reports that between a dense and a 75% weight-sparse checkpoint of the same model, exact component-to-component edge overlap falls to Jaccard@10 of 0.14 to 0.16, statistically indistinguishable from a random top-k baseline on one split ($p = 0.106$), while the coarser routing-head set overlaps at 0.55 to 0.67, far above its own baseline. The level at which circuits are compared can decide the answer.
- **Hidden mediator interactions.** The Curse of Multiple Mediators [7] re-derives the activation patching estimand from causal mediation analysis and shows that the natural indirect effect (NIE) decomposes as $NIE = PIE + INT$, where INT measures how much a component's causal effect depends on the state of other components. INT has been present in every activation patching result; components whose importance is conditional are either invisible or artificially inflated, and INT variance explains the previously documented instability of faithfulness scores. Compression perturbs the whole weight matrix at once, which is precisely the regime in which interaction terms move.

Research question. Does a circuit break under compression, or does the measurement break? Concretely: is the change in a task circuit under INT8-to-INT4 quantization or 10-to-60% weight pruning distinguishable from the change induced by a weight perturbation of matched magnitude and random direction, and does circuit-level damage agree with the feature-level damage already reported on the same models?

what has been audited under compression:

```
[+] SAE features      (Duan 2606.03002; Borobia 2603.25325)
[+] linear probes    (Duan 2606.15980)
[ ] CIRCUITS         <-- essentially nobody
    |
    v
but circuits inherit TWO measurement problems:
    |
    +-----+-----+
    |               |
    v               v
threshold-dependence  hidden mediator interactions
(CIRCUS, 2603.00523)  (NIE = PIE + INT, 2606.27510)
plus level-of-comparison  INT variance already explains
dependence (2607.18921)    faithfulness-score instability
    +-----+-----+
    |
    v
does a circuit "break" under compression, or does
the measurement break? the null model decides
```

2 Proposed Methodology

The ordering matters and is the methodological commitment of this project: the null model is built and characterised *before* any real compression is measured, so that the floor is not tuned after seeing the effect.

1. **Null model construction (first, not last).** For each compression setting we construct three matched controls. (a) *Matched-magnitude random perturbation*: perturb weights with the same per-tensor Frobenius change induced by INT-k rounding or by s% magnitude pruning, but in a random direction. (b) *Matched-perplexity control*: a different compression family tuned to the same perplexity, following the matched-perplexity pruning baseline used in [2]. (c) *Seed and threshold variation*: re-extraction under B non-nested pruning configurations, in the CIRCUS sense [5], and across extraction seeds. Together these give the distribution of circuit change attributable to nothing in particular.
2. **Circuit extraction under an explicitly reported protocol.** Attribution-graph construction followed by pruning, with the reported object, the pruning threshold, the query/key node representation, and the

comparison level all stated, per the reporting practice proposed in [6]. Two extraction algorithms are run in parallel (an attribution-patching graph and an edge-pruning graph) so that no conclusion rests on one pipeline.

3. **Stability-score reporting instead of binary membership.** Every edge carries an inclusion frequency $s(e)$ in $[0,1]$ over the configuration ensemble, and circuits are reported with the core / contingent / noise decomposition rather than as a single edge list. Change under compression is then a change in a distribution over edges, which is a quantity that can carry a confidence interval.
4. **Two-level comparison, always.** Every result is reported both as exact component-to-component edge overlap and as routing-head-set overlap, because [6] demonstrates these can disagree, with the former collapsing to chance while the latter remains well above it. A conclusion that holds at only one level is reported as such.
5. **Interaction-aware causal validation.** Where activation patching is used to confirm that a surviving or lost edge is causally real, we report the NIE alongside an interaction diagnostic: patching components singly versus in matched groups, so that a shift in NIE after compression is not read as a mechanism change when it is a change in the INT term [7]. Components flagged as interaction-dominated are reported separately and excluded from the headline stability claim.
6. **Cross-audit agreement test.** Duan [2] reports that RTN quantization and matched-perplexity magnitude pruning damage strongly overlapping SAE feature sets, with Jaccard overlap 0.79 to 0.86 and damage-score Spearman correlation 0.98. We compute the analogous circuit-level damage ranking on the same model and compression grid and test whether the two rankings agree. Disagreement is the informative outcome: it means a representation audit and a computation audit would license different deployment decisions.

2.1 Headline metric

Circuit Survival Index (CSI). For a compression setting c and a task circuit T , let $D(c)$ be the distance between the pre- and post-compression edge-inclusion-frequency vectors, and let $D_{\text{null}}(c)$ be the same distance under the matched-magnitude random perturbation. Then $\text{CSI}(c, T) = D(c) / D_{\text{null}}(c)$, reported with a bootstrap interval over the configuration ensemble and over seeds. CSI near 1 means compression damages the circuit no more than generic weight noise of the same size, and the observed “damage” is not evidence about compression at all. CSI substantially above 1 means compression is structurally selective, which is the claim that would justify a circuit-level audit in a deployment pipeline. CSI below 1 would indicate compression is *gentler* on circuits than random perturbation of the same magnitude, which is itself a publishable and somewhat surprising result.

3 Experimental Setup

3.1 Models

- **Primary:** Gemma-2-2B and Llama-3.2-1B. These are the models CIRCUS validates on [5] and are shared with the pruning-side feature audit [1], so the cross-audit agreement test in Section 2.6 runs on overlapping ground.
- **Secondary / cheap sweep:** Pythia-160M and Pythia-410M, for dense threshold sweeps and seed replication where the per-run cost of the primary models is prohibitive. Pythia-70M appears in [2] and provides a second point of contact with the feature-level results.
- **Canonical task circuits:** indirect object identification (the setting in which [7] demonstrates the interaction pathology), greater-than, and the docstring circuit. These are chosen because published reference circuits exist, so “change” is measured against something other than our own first extraction.

3.2 Compression grid

- **Quantization:** round-to-nearest at INT8 through INT4, matching the sweep in [2] so the bit-widths line up with the published feature-survival numbers, plus GPTQ and AWQ as error-compensated points.

- **Pruning:** unstructured magnitude and Wanda at 0 to 60% sparsity, matching the grid in [1]. Wanda is included specifically because [1] reports it preserves feature structure up to 3.7x better than magnitude pruning; whether that ordering also holds at the circuit level is a direct, pre-registered prediction to test.

3.3 Reported quantities

- CSI per (model, compression method, compression level, task circuit), at both comparison levels.
- Edge inclusion frequency distributions and the core / contingent / noise decomposition, pre- and post-compression.
- Rank correlation between circuit-level damage ordering and the feature-level damage ordering.
- Sensitivity of every headline conclusion to the pruning threshold, reported as an explicit sweep rather than a single operating point.
- Fraction of circuit components flagged interaction-dominated, and how that fraction itself moves under compression.

4 Measurement Protocol & Pre-Registration

Protocol rules fixed before any real-compression run

- 1. Null before effect.** The null distribution for every (model, compression level) cell is computed and frozen before the corresponding real-compression circuit is extracted. No null is re-tuned after seeing an effect.
- 2. Nothing binary.** No circuit is reported as an edge list. Every reported circuit is an inclusion-frequency vector over a fixed ensemble of non-nested pruning configurations.
- 3. Two levels or it does not count.** Any claim of the form “compression changed the circuit” must be stated at both the exact-edge and routing-head level, with the disagreement reported when it occurs.
- 4. Patching results carry an interaction flag.** No NIE is reported without the grouped-versus-single patching diagnostic attached.
- 5. Full protocol disclosure.** Reported graph object, extraction rule, threshold, Q/K node representation, and comparison level are stated for every figure, following the reporting practice of [6].

Algorithm 1 Null-Model Audit of Circuit Stability Under Compression

```

1: Input: model M, task T, compression settings C, pruning configs B, seeds S
2: -- Stage A: dense reference --
3: for b in B, s in S do
4:   G[b,s] <- ExtractCircuit(M, T, config=b, seed=s)
5: freq_dense(e) <- (1/|B||S|) * count of graphs containing edge e
6: core/contingent/noise <- Decompose(freq_dense)
7: -- Stage B: null distribution (FROZEN before Stage C) --
8: for c in C do
9:   delta(c) <- per-tensor Frobenius change induced by c on M
10:  for r in 1..R do
11:    M_null <- M + RandomPerturbation(magnitude=delta(c))
12:    freq_null[r] <- ExtractCircuitEnsemble(M_null, T, B, S)
13:    D_null(c) <- distribution over r of Dist(freq_dense, freq_null[r])
14: -- Stage C: real compression --
15: for c in C do
16:   M_c <- Compress(M, c)
17:   freq_c <- ExtractCircuitEnsemble(M_c, T, B, S)
18:   D(c) <- Dist(freq_dense, freq_c)
19:   CSI(c) <- D(c) / median(D_null(c)) [bootstrap CI over B, S, r]
20:   repeat lines 17-19 at routing-head granularity
21: -- Stage D: causal + cross-audit checks --
22: for each edge with large |freq_dense - freq_c| do
23:   NIE, INT_flag <- PatchSingle(e), PatchGrouped(e, siblings)
24:   if INT_flag then move e to interaction-dominated set (excluded from headline)
25: rank_circuit <- order C by D(c); rank_feature <- feature-damage order from [1,2]
26: report Spearman(rank_circuit, rank_feature) and threshold-sensitivity sweep

```

5 Compute Feasibility

Item	Plan
Hardware	DGX Spark, 128 GB unified memory. All target models are 2B or smaller and the workload is inference plus attribution, not training, which suits a high-memory / moderate-bandwidth box well. RTX 4060 handles the Pythia sweeps in parallel.
Dominant cost	The ensemble, not the model. Every cell is $ B $ configs $\times$ $ S $ seeds $\times$ R null draws. Budget is controlled by shrinking R and B , never by dropping the null. CIRCUS prunes one already-computed attribution run under many configurations, so the ensemble is close to free once the raw attribution exists [5].
Tooling	TransformerLens or nnsight for hooks and patching, SAELens for the dictionary-based extraction variant, and the public code releases accompanying [1], [5], and [6] as the starting point rather than a reimplementaion.
Scope control	If the grid overruns the two-month window, cut compression levels and secondary models first, then task circuits, then seeds. The null model and the two-level reporting are not cuttable; they are the contribution.

6 Principal Risks and Mitigations

- **The measurement basis is itself compressed.** A dictionary or attribution graph fitted on the dense model may drift for reasons unrelated to the circuit. Mitigation: run the dense-node extraction variant, which needs no learned dictionary, alongside the dictionary-based one, and report the divergence between them as a measured quantity rather than assuming it away. Duan's frozen-SAE-as-fixed-basis design [2] is the precedent for treating this explicitly.
- **The null is mis-specified.** Random perturbation of matched Frobenius magnitude may be a poor control for the structured error of rounding or of magnitude-based removal. Mitigation: three independent nulls (matched-magnitude, matched-perplexity, seed variation) with all three reported; if they disagree, that disagreement is a finding about how to audit compression, not a failure.
- **CSI turns out near 1 everywhere.** Then compression does not selectively damage circuits, which is a clean negative result and directly contradicts the qualitative claim in [4]. The stability-score protocol, the two-level reporting, and the cross-audit rank comparison all stand as contributions independently of which way CSI lands.
- **Someone publishes the LLM version first.** Plausible; [4] is already the VLM version and the area is moving fast. Mitigation: the differentiator is the null model and the pre-registered protocol, not the observation. A paper reporting circuit change without a floor does not close this gap, it widens it.

7 Expected Contributions for AAI UC

1. **The first circuit-level compression audit for language models, with a floor.** Existing audits cover features and probes; this covers computation, and it is the first to report circuit change against a matched-magnitude null rather than against nothing.
2. **A reusable stability protocol.** Circuit Survival Index, inclusion-frequency reporting, mandatory two-level comparison, and the interaction-flagged patching diagnostic, packaged so that any future “did X change the circuit?” question (fine-tuning, distillation, RLHF, LoRA merging) can be asked the same way.
3. **A direct test of whether representation audits and computation audits agree.** If the feature-level and circuit-level damage rankings correlate, the cheaper feature audit is a valid proxy and deployment pipelines can use it. If they do not, that is a concrete warning about a method the field is currently adopting.

4. **Open tooling.** Release of the null-generation harness, the ensemble extraction wrapper, and the full CSI grids, so that the floor does not have to be rebuilt by the next group.

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Implementation Roadmap

Phase	Milestones & Tasks
Phase 1	Infrastructure & Dense Reference – Set up extraction stack (TransformerLens / nnsight, SAELens) and reproduce a published reference circuit on the dense model as a sanity anchor. – Implement the CIRCUS-style ensemble wrapper: B non-nested pruning configurations, edge inclusion frequencies, core / contingent / noise decomposition. – Fix and document the reporting protocol (graph object, threshold, Q/K representation, comparison level) before any compression run.
Phase 2	Null Model Construction and Freezing – Implement matched-magnitude random perturbation, matched-perplexity control, and seed variation. – Characterise the null distribution of circuit change for every compression cell on the Pythia sweep first, then the primary models. – Freeze the nulls. This is the pre-registration point; nothing downstream may re-tune them.
Phase 3	Real Compression Measurement – Run the quantization grid (RTN INT8 to INT4, plus GPTQ / AWQ) and the pruning grid (magnitude and Wanda, 0 to 60%). – Compute CSI at both the exact-edge and routing-head levels, with bootstrap intervals. – Test the pre-registered prediction from the feature-level literature: does Wanda preserve circuits better than magnitude pruning, as it does features?
Phase 4	Causal Validation and Cross-Audit – Activation-patch the largest inclusion-frequency movers; attach the grouped-versus-single interaction diagnostic and split out interaction-dominated components. – Compute the circuit-level damage ranking and correlate it against the published feature-level damage ranking on the same models and compression grid. – Run the full threshold-sensitivity sweep and report which conclusions survive it.
Phase 5	AAAI UC Paper Preparation – Produce the headline CSI figures, the inclusion-frequency shift plots, and the level-of-comparison disagreement figure. – Draft and format for AAAI UC; prepare poster and the open-source null-generation harness. – Write the limitations section around the three named risks rather than around a generic future-work paragraph.